

Sign of Permutations in S_2

$$\text{Inv}(\pi) = \{(i, j) : i < j \text{ and } \pi(i) > \pi(j)\}$$

For example,

$$\text{Inv}(\pi_1) = \emptyset \quad \begin{array}{l} \text{no inversions} \\ \text{for permutation} \\ \pi_1 \\ \Rightarrow n(\pi_1) = 0 \end{array}$$

Consider pair $(1, 2)$

we want to know

Input: $1 < 2$ *what is the relation?*

$$\begin{array}{ccc} \text{Output: } \pi_1(1) & R & \pi_1(2) \\ || & & || \\ 1 & < & 2 \end{array}$$

Because $\pi(i) < \pi(j)$, there are no inversions for π_1

Let's check π_2 :

$$\text{Inv}(\pi_2) = \{(1, 2)\}$$

$(1, 2)$ is an inversion
w/r to π_2

Consider $(1, 2)$

$$\begin{array}{ccc} \text{Input: } 1 & < & 2 \\ \text{Output: } \pi_2(1) & R & \pi_2(2) \\ || & & || \\ 2 & > & 1 \end{array}$$

*note: opposite in input
and output!*

$$\text{sgn}(\pi) = (-1)^{n(\pi)} \quad \leftarrow \text{\# of inversions}$$

$$\Rightarrow \text{sgn}(\pi_1) = (-1)^{n(\pi_1)} = (-1)^0 = \boxed{1} \quad \text{even \# of transpositions (inversions)}$$

$$\text{sgn}(\pi_2) = (-1)^{n(\pi_2)} = (-1)^1 = \boxed{-1} \quad \text{odd \# of transpositions (inversions)}$$